WORKED SOLUTION

STEP II 2007, Question 11

Mechanics • Projectiles in Three Dimensions

STEP II 2007, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q11

The key idea

Write the motion as a single position vector: the horizontal velocity is constant, and only the vertical component feels gravity. Once $\mathbf{r}(t)$ is down on paper, the distance from O is just $|\mathbf{r}|$, and the shortest distance comes from minimising a quadratic. The last part works because the bullet is so fast that the particle barely moves while it is in flight.

Solution

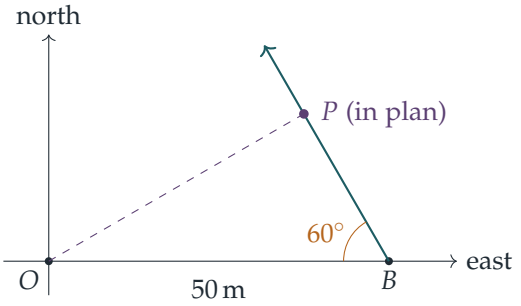
We model the particle as a point moving under gravity alone ($g = 10\,\mathrm{m\,s^{-2}}$, no air resistance), and take O as the origin, so B has position vector $50\mathbf{i}$.

Part (i)

Components of the initial velocity. If $\tan\alpha = \frac{1}{2}$ then $\cos\alpha = \frac{2}{\sqrt{5}}$ and $\sin\alpha = \frac{1}{\sqrt{5}}$ (a 1, 2, $\sqrt{5}$ triangle). So the initial velocity has

$$\text{horizontal speed } 25 \cdot \tfrac{2}{\sqrt{5}} = 10\sqrt{5}, \qquad \text{vertical speed } 25 \cdot \tfrac{1}{\sqrt{5}} = 5\sqrt{5}.$$

Which horizontal direction? The direction makes 60° with the line OB. If the particle headed east-ish, away from O, it could never pass O at all; to pass to the north of O it must head back towards the west and northwards. So the horizontal direction is at 60° to $\overrightarrow{BO}$ (west), on the north side, giving the unit vector $-\frac{1}{2}\mathbf{i} + \frac{\sqrt{3}}{2}\mathbf{j}$.



The horizontal velocity is therefore

$$10\sqrt{5}\left(-\tfrac{1}{2}\mathbf{i}+\tfrac{\sqrt{3}}{2}\mathbf{j}\right)=-5\sqrt{5}\,\mathbf{i}+5\sqrt{15}\,\mathbf{j},$$

and it stays constant, since gravity acts vertically. Vertically, $z=5\sqrt{5}\,t-\frac{1}{2}(10)t^2$. Adding the starting point $50\mathbf{i}$,

$$\mathbf{r}(t)=(50-5\sqrt{5}\,t)\mathbf{i}+5\sqrt{15}\,t\,\mathbf{j}+(5\sqrt{5}\,t-5t^2)\mathbf{k}.$$

The distance from O. Squaring and adding the components,

$$\begin{aligned}|\mathbf{r}|^2&=(2500-500\sqrt{5}\,t+125t^2)+375t^2+(125t^2-50\sqrt{5}\,t^3+25t^4)\\&=25t^4-50\sqrt{5}\,t^3+625t^2-500\sqrt{5}\,t+2500.\end{aligned}$$

Compare this with

$$\begin{aligned}25(t^2-\sqrt{5}\,t+10)^2&=25(t^4+5t^2+100-2\sqrt{5}\,t^3+20t^2-20\sqrt{5}\,t)\\&=25t^4-50\sqrt{5}\,t^3+625t^2-500\sqrt{5}\,t+2500,\end{aligned}$$

which is the same. So $|\mathbf{r}|=5\,|t^2-\sqrt{5}\,t+10|$. The quadratic $t^2-\sqrt{5}\,t+10$ has discriminant $5-40<0$, so it is always positive and we can drop the modulus:

$$|\mathbf{r}|=5(t^2-\sqrt{5}\,t+10)\text{ m},$$

as required.

The point P. Completing the square,

$$t^2-\sqrt{5}\,t+10=(t-\tfrac{\sqrt{5}}{2})^2+\tfrac{35}{4},$$

so the distance is least when $t=\frac{\sqrt{5}}{2}$, and the shortest distance is $5\cdot\frac{35}{4}=\frac{175}{4}=43.75\,\text{m}$. Substituting $t=\frac{\sqrt{5}}{2}$, so that $\sqrt{5}\,t=\frac{5}{2}$ and $t^2=\frac{5}{4}$:

$$\overrightarrow{OP}=\tfrac{75}{2}\,\mathbf{i}+\tfrac{25\sqrt{3}}{2}\,\mathbf{j}+\tfrac{25}{4}\,\mathbf{k}\quad\text{(in metres)}.$$

The horizontal part of $\overrightarrow{OP}$ has east component $\frac{75}{2}$ and north component $\frac{25\sqrt{3}}{2}$. A bearing θ is measured clockwise from north, so

$$\tan\theta=\frac{\text{east}}{\text{north}}=\frac{75/2}{25\sqrt{3}/2}=\sqrt{3},$$

and since both components are positive, $\theta=60^\circ$: the bearing of P from O is 060°.

Part (ii)

The vertical component of velocity is

$$\dot{z} = 5\sqrt{5} - 10t,$$

which is zero exactly when $t = \frac{\sqrt{5}}{2}$. This is the same time at which the particle is at P, so the particle is at its highest point (height $\frac{25}{4}$ m) at P.

Part (iii)

The bullet travels in a straight line from O to P at $350\,\mathrm{m\,s^{-1}}$ (we ignore gravity on it), covering $OP = \frac{175}{4}$ m. It takes

$$T = \frac{175/4}{350} = \frac{1}{8}\ \mathrm{s}.$$

? **Ask yourself** Where is the particle after this extra $\frac{1}{8}$ second?

By part (ii), at P the particle's vertical velocity is zero, so its velocity there is purely horizontal, of magnitude $10\sqrt{5}\,\mathrm{m\,s^{-1}}$. In the next T seconds it moves horizontally $10\sqrt{5}\,T$ and drops $\frac{1}{2}gT^2$. With $T = \frac{1}{8}$,

$$\text{horizontal shift} = \frac{10\sqrt{5}}{8} \approx 2.80\,\mathrm{m}, \qquad \text{drop} = 5 \cdot \frac{1}{64} \approx 0.08\,\mathrm{m}.$$

These are at right angles, so the distance from P is

$$\sqrt{\frac{500}{64} + \frac{25}{4096}} = \frac{5\sqrt{1281}}{64} \approx 2.80\,\mathrm{m},$$

which is approximately 3 m, as required. (The drop hardly matters: the answer is essentially $10\sqrt{5}/8$.) The marksman misses.

Checks

- At $t = 0$ the formula gives $5 \cdot 10 = 50$ m, the distance OB. Good.
- $|\overrightarrow{OP}|^2 = 37.5^2 + 21.65^2 + 6.25^2 = 1406.25 + 468.75 + 39.06 = 1914.06 = 43.75^2$, so the position vector of P agrees with the minimum distance.
- A numerical scan of $|\mathbf{r}(t)|$ gives a minimum of 43.75 m at $t \approx 1.118 = \frac{\sqrt{5}}{2}$, and stepping the particle on from P for 0.125 s gives a separation of 2.796 m.

Common slips

- Taking the horizontal direction as $\frac{1}{2}\mathbf{i} + \frac{\sqrt{3}}{2}\mathbf{j}$ (away from O). Then the particle never passes north of O, and the distance does not come out as given.

- Giving the bearing as $30°$: that is the angle from east, but bearings are measured clockwise from north.

Beyond STEP

Part (iii) is a baby version of the interception problem: to hit a moving target you must aim where it *will* be, which leads to solving $|\mathbf{r}_{\text{target}}(t) - \mathbf{r}_{\text{bullet}}(t)| = 0$ for both the firing direction and the time. Minimising $|\mathbf{r}(t)|$ is also the idea behind "distance of closest approach" in orbital and collision problems, where squaring the distance to get a polynomial is the standard first move. Kleppner and Kolenkow's *An Introduction to Mechanics* develops vector kinematics of this kind properly.

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